2.6 The Arithmetic–Geometric Mean

WORKSHEET

The question

STEP II 2024, Question 8

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2024 STEP papers (a zip file):

www.ocr.org.uk/Images/725753-2024-past-papers.zip

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps. There is space after each hint for your working.

Ask yourself Why is the arithmetic mean of two different positive numbers always bigger than their geometric mean? Can you prove it in one line?

STAGE 1 The two sequences squeeze together

Covers: Part (i)

Ask yourself Use the hint in the question: $(\sqrt{x_n} - \sqrt{y_n})^2 > 0$. What does it say about x_{n+1} and y_{n+1}?

Ask yourself Why is $x_{n+1} < x_n$ and $y_{n+1} > y_n$? What does that tell you about the two sequences?

Pause and try For the gap, compare $x_{n+1} - y_{n+1}$ with $x_{n+1} - y_n$. By what factor does the gap shrink at each step?

STAGE 2 The substitution

Covers: Part (ii)

Ask yourself With $t = \frac{1}{2}\left(x - \frac{pq}{x}\right)$, compute $t^2 + pq$ and $t^2 + \frac{1}{4}(p+q)^2$. Do they factorise?

Pause and try Also find $\frac{\mathrm{d}t}{\mathrm{d}x}$. As x runs from 0 to ∞, what does t do? Put the pieces together, and use the fact that the t-integrand is even.

STAGE 3 **Evaluating the integral**

Covers: Part (ii)

Pause and try Apply the identity repeatedly, starting from (x_0, y_0). What happens to $\mathrm{I}(x_n, y_n)$ as $n \to \infty$? Evaluate $\mathrm{I}(M, M)$ directly.

What's really going on

Faster than the question admits

Part (i) shows the gap at least halves at each step, which would give about one extra binary digit per step. The table in the hook shows something far more dramatic: the number of correct digits roughly *doubles* at each step $(2, 4, 10, 21, 43, \ldots)$. In fact

$$x_{n+1} - y_{n+1} = \frac{1}{2}(\sqrt{x_n} - \sqrt{y_n})^2 = \frac{(x_n - y_n)^2}{2(\sqrt{x_n} + \sqrt{y_n})^2},$$

so each gap is roughly a constant times the *square* of the previous one. This is called *quadratic convergence*, and we will study it properly in Section 2.7.

Elliptic integrals

Our integral $\mathrm{I}(p, q)$ is a disguised *complete elliptic integral*, a type of integral that arises in computing the arc length of an ellipse and that cannot be expressed using the functions you know. Its most famous application is the pendulum. The A-level formula $T = 2\pi\sqrt{L/g}$ for the period is only an approximation for small swings. The exact period of a pendulum released from angle θ_0 is

$$T = \frac{2\pi\sqrt{L/g}}{M(1, \cos\frac{\theta_0}{2})},$$

an elliptic integral evaluated by the AGM. For $\theta_0 = 90^\circ$ the true period is about 18% longer than the simple formula predicts. Gauss's diary entry concerned the lemniscate, a figure-of-eight curve, whose total arc length involves $\int_0^1 \frac{\mathrm{d}t}{\sqrt{1-t^4}}$; that integral equals $\mathrm{I}(\sqrt{2}, 1)$ in disguise, so the AGM evaluates it.

Computing π

In 1976 Richard Brent and Eugene Salamin independently turned Gauss's ideas into an algorithm for computing π that doubles the number of correct digits at every step. Twenty-five steps give tens of millions of digits. Algorithms of this kind were used in record-breaking computations of π in the 1980s and 1990s, and quadratically convergent algorithms like it are now standard wherever high precision is needed. Compare Viète's product in Section 2.5, which gains about 0.6 of a digit per step.

Try this next

1 **Quadratic convergence.** Prove the identity $x_{n+1} - y_{n+1} = \dfrac{(x_n - y_n)^2}{2(\sqrt{x_n} + \sqrt{y_n})^2}$, and deduce that $x_{n+1} - y_{n+1} \le \dfrac{(x_n - y_n)^2}{8y_0}$.

2 **By hand.** Compute $M(2, 1)$ to six decimal places using a calculator. How many steps did you need?

3 **Scaling.** Show that $M(\lambda x, \lambda y) = \lambda M(x, y)$ for $\lambda > 0$, and that $\mathrm{I}(\lambda p, \lambda q) = \lambda^{-1}\mathrm{I}(p, q)$. Check that these two facts are consistent with $\mathrm{I}(x_0, y_0) = \pi/(2M)$.

4 **The pendulum.** Use the formula above to find, to three significant figures, how much longer the period of a pendulum released from 60° is than the small-angle prediction.

Further reading

- **David Cox, "The arithmetic–geometric mean of Gauss"**, *L'Enseignement Mathématique* 30 (1984). The classic account of the history and mathematics; the first sections need only calculus.
- **Jonathan Borwein & Peter Borwein, *Pi and the AGM*** (Wiley). The definitive book on the AGM and fast computation of π.
- **Jonathan Borwein & Keith Devlin, *The Computer as Crucible*** (A. K. Peters). Experimental mathematics, including discoveries like Gauss's, made by spotting numerical coincidences.

The full guided solution is in Section 2.6:
https://beyondstep.keithmcnulty.org/ch02/agm.html